

Lecture 12: Loss of Significant Digits; Stability

Math 6610

C. Mou

Department of Mathematics and Statistics
Utah State University

Fall 2026

Today's Goals

- Avoid cancellation by rewriting a formula: multiply by the conjugate, split the domain
- Recognize errors caused by very large intermediate values
- Define stability of an algorithm $\tilde{F}$: nearly the right answer to nearly the right problem

Loss of Significant Digits: Rewriting

Subtraction of nearly equal numbers loses digits (Lecture 11):

$$x - y, \quad x \approx y.$$

Often an equivalent formula has no such subtraction.

Example (*Multiply by $1 + \cos x$*). Evaluate

$$f(x) = \frac{1 - \cos x}{\sin^2 x} \quad \text{for } 0 < x < \pi.$$

For x near 0, $\cos x \approx 1$. Multiply numerator and denominator by $1 + \cos x (> 0$ on $0 < x < \pi$) and use $1 - \cos^2 x = \sin^2 x$:

$$f(x) = \frac{1 - \cos x}{\sin^2 x} \cdot \frac{1 + \cos x}{1 + \cos x} = \frac{1 - \cos^2 x}{\sin^2 x} \cdot \frac{1}{1 + \cos x} = \frac{1}{1 + \cos x}.$$

Loss of Significant Digits: Splitting the Domain

Example (continued). The new formula has its own problem: for x near π , $\cos x \approx -1$, and $1 + \cos x$ subtracts nearly equal numbers. Use each formula where it is safe:

$$f(x) = \begin{cases} \frac{1}{1 + \cos x} & 0 < x \leq \frac{\pi}{2} \\ \frac{1 - \cos x}{\sin^2 x} & \frac{\pi}{2} < x < \pi \end{cases}$$

Remark (Why both parts are safe). For $0 < x \leq \pi/2$: $\cos x \geq 0$, so $1 + \cos x$ adds two numbers ≥ 0 . For $\pi/2 < x < \pi$: $\cos x < 0$, so $1 - \cos x$ adds two positive numbers. Neither case subtracts nearly equal numbers.

Large Intermediate Values: Example (1/2)

Example (*A formula with a large intermediate value*).

$$f(x) = \frac{1}{x} \left(\frac{1}{x} \frac{1}{\frac{1}{x} - 1} - 1 \right), \quad 0 < x < 1.$$

Evaluate at $x = 2^{-54}$. In exact arithmetic:

$$y = f(2^{-54}) = 2^{54} \left(\frac{2^{54}}{2^{54} - 1} - 1 \right) = \frac{1}{1 - 2^{-54}} \approx 1.$$

(Check: $\frac{2^{54}}{2^{54} - 1} - 1 = \frac{1}{2^{54} - 1}$; multiply by 2^{54} .)

Large Intermediate Values: Example (2/2)

In floating-point arithmetic:

$$\begin{aligned}\tilde{y} &= 2^{54} \otimes \left((2^{54} \oslash (2^{54} \ominus 1)) \ominus 1 \right) \\ &= 2^{54} \otimes \left((2^{54} \oslash 2^{54}) \ominus 1 \right) \\ &= 2^{54} \otimes (1 \ominus 1) = 2^{54} \otimes 0 = 0.\end{aligned}$$

First step: doubles near 2^{54} are $2^{54}\epsilon_{\text{mach}} = 4$ apart, so $2^{54} - 1$ rounds to 2^{54} .

$$\text{Relative error} = \left| \frac{\tilde{y} - y}{y} \right| = 1 \quad (100\%).$$

The cause is the intermediate value 2^{54} , much larger than the input 2^{-54} and the output ≈ 1 .

Large Intermediate Values: Solution

Simplify before computing. Since $\frac{1}{x} \cdot \frac{1}{\frac{1}{x}-1} = \frac{1}{1-x}$,

$$f(x) = \frac{1}{x} \left(\frac{1}{1-x} - 1 \right) = \frac{1}{x} \cdot \frac{x}{1-x} = \frac{1}{1-x}.$$

In floating-point arithmetic:

$$\tilde{y} = 1 \oslash (1 \ominus 2^{-54}) = 1 \oslash 1 = 1.$$

$(1 - 2^{-54})$ lies halfway between the doubles $1 - 2^{-53}$ and 1; the tie goes to 1.)

Relative error:

$$\left| \frac{\tilde{y} - y}{y} \right| = \left| \frac{1 - \frac{1}{1-2^{-54}}}{\frac{1}{1-2^{-54}}} \right| = |(1 - 2^{-54}) - 1| = 2^{-54} < \frac{1}{2} \epsilon_{\text{mach}}.$$

Why the First Formula Fails

Remark (*Each step alone is accurate*). $2^{54} \ominus 1 = 2^{54}$ has relative error 2^{-54} , within the bound of Lecture 11. The lost unit is small compared with 2^{54} .

Remark (*The steps together are not*). The exact value of $\frac{2^{54}}{2^{54}-1} - 1$ is $\frac{1}{2^{54}-1} \approx 2^{-54}$, and it comes entirely from that lost unit. The computed value is 0. Multiplying by 2^{54} keeps the relative error at 1.

Remark (*Rule*). Intermediate values much larger than the input and the output indicate a possible loss of accuracy. Simplify the formula first.

Summary: Ways to Avoid Cancellation

- ❶ **Multiply by the conjugate.** $1 - \cos x = \frac{\sin^2 x}{1 + \cos x},$

$$\sqrt{x+1} - \sqrt{x} = \frac{1}{\sqrt{x+1} + \sqrt{x}} \quad (x \geq 0).$$

A subtraction is replaced by an addition.

- ❷ **Split the domain.** Use each formula only where it has no cancellation.
- ❸ **Use a series near the bad point.** For small $|x|,$

$$1 - \cos x = \frac{x^2}{2} - \frac{x^4}{24} + \cdots,$$

with error after the x^4 term at most $x^6/720$ (alternating series with decreasing terms, $|x| \leq 1$).

- ❹ **Simplify algebraically first,** e.g. $f(x) = \frac{1}{1-x}$ above.

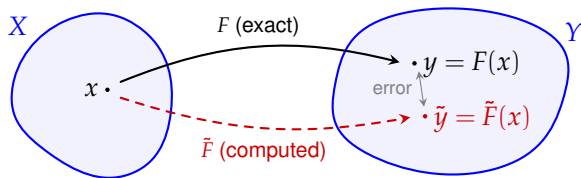
Problems and Algorithms (1/2)

Definition (*Problem, algorithm*). A *problem* is a map $F : X \rightarrow Y$ between normed vector spaces (Lecture 9): input $x \in X$, output $y = F(x) \in Y$. An *algorithm* for F is a map $\tilde{F} : X \rightarrow Y$ computed in floating-point arithmetic. $\tilde{F}(x)$ includes all errors:

- rounding the input, $x \rightarrow \text{fl}(x)$
- rounding in each floating-point operation
- approximation errors (e.g. stopping Newton's method after finitely many steps)

Problems and Algorithms (2/2)

Exact solution: $y = F(x)$. Computed solution: $\tilde{y} = \tilde{F}(x)$.



Accuracy and Its Limit

Definition (Accurate algorithm). $\tilde{F}$ is *accurate* if, for all $x \in X$ with $F(x) \neq 0$,

$$\frac{\|\tilde{y} - y\|}{\|y\|} = \frac{\|\tilde{F}(x) - F(x)\|}{\|F(x)\|} = O(\epsilon_{\text{mach}}). \quad (1)$$

Remark (Meaning of $O(\epsilon_{\text{mach}})$). $g(\epsilon) = O(\epsilon)$: there are constants $C, \epsilon_0 > 0$ with $\|g(\epsilon)\| \leq C|\epsilon|$ for all $|\epsilon| \leq \epsilon_0$. Here C must not depend on x .

Remark (Limit). Rounding the input alone changes x by a relative amount up to $\frac{1}{2}\epsilon_{\text{mach}}$, which changes $F(x)$ by a relative amount up to about $\kappa(x) \frac{1}{2}\epsilon_{\text{mach}}$ (condition number, Lecture 9). So (1) is possible when $\kappa(x)$ is of moderate size, but cannot be expected when $\kappa(x) \gg 1$ (ill-conditioned problem).

Stability: Definition

Instead of (1), we ask for less:

Definition (Stable). The algorithm $\tilde{F} : X \rightarrow Y$ for the problem F is *stable* if for each $x \in X$, $x \neq 0$, there exists $\tilde{x} \in X$ such that

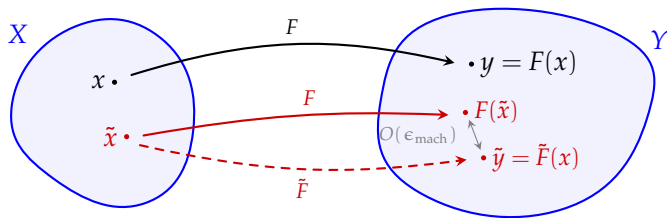
$$(1) \quad \frac{\|\tilde{F}(x) - F(\tilde{x})\|}{\|F(\tilde{x})\|} = O(\epsilon_{\text{mach}}),$$

$$(2) \quad \frac{\|\tilde{x} - x\|}{\|x\|} = O(\epsilon_{\text{mach}}),$$

with constants in $O(\epsilon_{\text{mach}})$ that do not depend on x . (In (1), read the quotient as 0 if $\tilde{F}(x) = F(\tilde{x}) = 0$.)

Remark (Meaning). A stable algorithm gives nearly the right solution to nearly the right problem.

Stability: The Picture



Remark (What stability says). The computed $\tilde{y}$ is close to $F(\tilde{x})$, and $\tilde{x}$ is close to x , both in relative size $O(\epsilon_{\text{mach}})$. Any larger difference between $\tilde{y}$ and $y = F(x)$ comes from the sensitivity of the problem ($\kappa(x)$), not from the algorithm. This is a realistic goal.

Remark. Next lecture: *backward stability*, where $\tilde{F}(x) = F(\tilde{x})$ holds exactly.

Looking Ahead

- **Next lecture:** Backward Stability